

# The Date Crew: Product & Tech Generalist Assessment

**Prototype (live):** <https://jatanrajbhar.github.io/match-check/> · **Code:** [github.com/jatanrajbhar/match-check](https://github.com/jatanrajbhar/match-check) · Every number below is reproduced by `analysis/funnel_analysis.py` (see the appendix for the full table and assumptions).

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## Part 1: Diagnose the problem

### 1. Three questions I would investigate

**a. Which profiles get rejected, and why?** 690 of 1,000 profiles (69%) are rejected, the largest drop in the funnel. Feedback is free text, so nobody can say which reasons dominate and every fix is a guess. Structured reasons would also show whether the 35% "already in preferences" rejections cluster on a few fields.

**b. Are the same profiles circulated to many clients?** Over-shared profiles get fatigued and decline more, which may explain part of the 310 → 210 drop after clients say yes (assuming that step needs mutual consent). It is also unfair to less-exposed members.

**c. Do matchmakers recommend the same "type" to every client?** A converts 44%, B 21%. If A tailors and B sends a template type, part of the fix is coaching, not software (after controlling for client mix). The numbers imply B sends *more* profiles than A ( $\approx 565$  vs  $\approx 435$ ).

### 2. The biggest problem

The quality and relevance of what we send at the top of the funnel, caused by the absence of a learning loop. Profiles are shared without being checked against stated preferences, and rejection feedback is not captured in a form that improves the next recommendation.

In numbers: 35% of 690  $\approx$  **241 rejections a month, 24% of everything we send, were for something the client had already told us.**

*Assumptions:* the counts are one 30-day cohort; the 35% applies to all rejections; later steps convert at 50–71%, so fixes at the top multiply through them.

### 3. Three metrics

- Meetings completed per matchmaker-hour** (north star). Hours aren't logged yet, so instrument them. Assuming 100 clients at 2 h/week:  $\approx 857$  search hours for 42 meetings.
- Avoidable rejection rate:** rejections for a reason already in stated preferences  $\div$  profiles shared. Today  $\approx 24\%$ . A leading indicator that moves within days.
- Meeting completion rate:** completed  $\div$  fixed, today 56%. A guardrail that gains reach real meetings; segment no-shows and cancellations by reason.

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## Part 2: Design a solution

### Problem

One in four profiles we send ( $\approx 241$  a month) is rejected for a reason the client already gave us: a stated preference or a deal-breaker. Preferences are not checked at the moment of sending, and free-text feedback never becomes something the next search can use.

## User

Matchmakers, at the moment they are about to email a profile. Secondly the matchmaking lead, who sees the avoidable-rejection rate per matchmaker each week. Clients never see the tool.

## Solution: "Match Check"

1. **Pre-send check (rules, not AI).** Each candidate is compared with the client's stated preferences and deal-breakers. **Blocked** if it breaks one (sending anyway needs a logged reason); **Review** if a field is missing, a deal-breaker is borderline ("open to kids" vs a firm yes) or it matches a learned pattern; otherwise **Clear**. Plain code is explainable, instant and free.
2. **Feedback structuring (AI).** A client's reply becomes structured reasons from a fixed 16-category taxonomy, each with a quote, firmness (firm/soft) and direction ("wants taller"). Code, not the model, then decides if the rejection was avoidable.
3. **Learned signals.** Rejections on checkable fields the stated preferences don't cover become signals, e.g. "rejected 2 of 2 profiles 5'6" or shorter". They only flag for review, never block, and show counter-evidence, because some clients reject a profile and later accept a very similar one. A consistent signal prompts the matchmaker to confirm with the client and add it to the stated preferences, closing the loop.

**Two-week plan.** Days 1–3: import preferences, profiles and the sent log into one table and normalise fields. Days 4–6: rules engine with tests, taxonomy and prompt, backfill 90 days of feedback for a baseline. Days 7–10: internal page (shortlist, reply logging, overrides) and weekly report. Days 11–14: pilot with Matchmaker B (most volume, lowest acceptance) and hand-check 50 parsed replies.

## Data

Client preferences and deal-breakers (structured), candidate profiles with the same fields, the sent log (client, candidate, matchmaker, date), reply text with accept/reject, and later funnel outcomes. Missing fields return "can't verify", never a silent pass.

## Technology

Rules in JavaScript (the prototype runs in the browser). Claude Opus 5.5 via the Anthropic API with JSON-schema structured output, so every answer fits the taxonomy; at roughly 1.5k input and 0.5k output tokens per reply, 700 rejections a month cost about \$11. The Message Batches API (half price) for backfills. Postgres or Google Sheets for storage at first, and later the Gmail API to pull replies automatically.

## Success metric

**Avoidable rejection rate falls from  $\approx 24\%$  of profiles shared to under 5% within four weeks.** If the freed slots are refilled with profiles that accept at today's non-avoidable rate (40.9%), acceptance rises from 31% to  $\approx 41\%$  and completed meetings from 42 to  $\approx 55$  a month ( $\approx 52$  if replacements only match today's average). Guardrails: fewer than 10% of blocks overridden, profiles shared per client does not drop, search time per client does not rise.

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## Part 3: Prototype

**Match Check** is a single-page internal tool. Live: <https://jatanrajbhar.github.io/match-check/> · Code: [github.com/jatanrajbhar/match-check](https://github.com/jatanrajbhar/match-check)

- **Pre-send check.** Pick a client and every unsent profile in the pool is marked Clear, Review or Blocked, with the reason and its source (stated preference, deal-breaker, learned, missing data).
- **Feedback structuring.** Paste a client's reply and it becomes structured reasons, with a verdict on whether the rejection was avoidable. Uses **Claude** when you enter an Anthropic API key, or offline

keyword rules without one, so the demo always works.

- **Learning loop.** Saving a reply updates that client's learned signals, the shortlist and the team metrics immediately.
- **Team metrics.** Acceptance and avoidable-rejection rate per matchmaker. The 25 mock replies mirror the brief: A 45% vs B 21%, and 35% of rejections avoidable.

What is real and what is mocked: the rules engine, avoidability logic, learned signals and metrics are real code covered by 10 unit tests ( `node --test` ). The Claude call is real SDK code; with an invalid key it returns a handled 401, confirming the request path, but I have not run it with a live key. Clients, profiles and replies are mock data.

Match Check

Feedback parserOffline keyword rulesClaude (claude-opus-5-5)

Checks a profile against the client's stated preferences *before* it is emailed, and turns free-text rejection feedback into structured reasons. Prototype · mock data.

25

Profiles shared

last 30 days

32%

Acceptance rate

8 accepted

24%

Avoidable rejection rate

6 of 25 shared profiles were rejected for a reason the client had already stated (35% of rejections)

|              | Shared | Accepted | Rate | Avoidable |
|--------------|--------|----------|------|-----------|
| Matchmaker A | 11     | 5        | 45%  | 2 (18%)   |
| Matchmaker B | 14     | 3        | 21%  | 4 (29%)   |

Clients

Priya Sharma

29 · Mumbai · Matchmaker B

2/7 profiles accepted

Sneha Iyer

27 · Chennai · Matchmaker B

1/7 profiles accepted

Aman Verma

33 · Bangalore · Matchmaker A

3/6 profiles accepted

Kabir Khan

31 · Delhi · Matchmaker A

2/5 profiles accepted

STATED PREFERENCES

Age29–35

Height5'9"+ (175 cm)

LocationMumbai, Pune (or willing to relocate)

ReligionHindu

EducationMasters or higher

Marital statusNever married

DietAny

DEAL-BREAKERS

Smokingup to "Never"

Drinkingup to "Socially"

Wants childrenYes

LEARNED FROM FEEDBACK

Rejections on things the stated preferences don't cover. These only ever flag a profile for review, never block it.

Mixed signal

Location: Pune. Rejected 1, accepted 1 similar.

"Seems nice, but Pune feels too far honestly."

OTHER THEMES (CAN'T BE CHECKED AUTOMATICALLY)

Photos / attraction × 1

Career / ambition × 1

Pre-send check · 8 profiles in the pool

CLEAR TO SEND

Ishaan Kapoor

· Entrepreneur

32 · 5'11" · Mumbai · Hindu · Masters · Never married · Non-vegetarian · Smokes: Never · Drinks: Socially · Kids: Yes

Meets every stated preference and deal-breaker.

Log client reply

CLEAR TO SEND

Vivek Raman

· Product designer

31 · 5'11" · Chennai (will relocate) · Hindu · Masters · Never married · Vegetarian · Smokes: Never · Drinks: Never · Kids: Yes

Meets every stated preference and deal-breaker.

Log client reply

REVIEW

Nitin Agarwal

· Chartered accountant

31 · height ? · Mumbai · Hindu · Masters · Never married · Vegetarian · Smokes: Never · Drinks: Socially · Kids: Yes

MISSING DATA Height not on profile, can't verify

Log client reply

BLOCKED

Dev Malhotra · Lawyer

34 · 6'0" · Mumbai · Hindu · Masters · Never married · Non-vegetarian · Smokes: Occasionally · Drinks: Socially · Kids: Yes

DEAL-BREAKER Smoking: Occasionally (client accepts up to "Never")

Log client reply

REVIEW

Manish Joshi · Software architect

30 · 5'10" · Pune · Hindu · Masters · Never married · Vegetarian · Smokes: Never · Drinks: Socially · Kids: Yes

LEARNED Rejected 1 of 2 similar profiles on location (Pune). Mixed signal: client has also accepted 1, so treat as a soft preference

Log client reply

BLOCKED

Abhishek Nair · Doctor

29 · 5'6" · Chennai · Hindu · Masters · Never married · Vegetarian · Smokes: Never · Drinks: Never · Kids: Yes

STATED PREFERENCE Height 5'6" (167 cm) is outside the stated range

STATED PREFERENCE Lives in Chennai, not willing to relocate (client wants Mumbai / Pune)

Log client reply

Log a client reply

Profile

Dev Malhotra (34, Mumbai)

☒ Rejected
☐ Accepted

He smokes! I was very clear that I don't want a smoker. Also he's 34, a bit older than I'd like.

Try: He smokes, and I said no smokers. Please read my preferences.

Seems lovely but lives too far away, I'd rather someone in my city.

Honestly not sure, the photos didn't really do it for me.

Structure feedback

STRUCTURED REASONS

via offline keyword rules

| Reason  | Client said                                | Firmness | Direction   |
|---------|--------------------------------------------|----------|-------------|
| Smoking | "He smokes!"                               | firm     | –           |
| Age     | "Also he's 34, a bit older than I'd like." | soft     | wants lower |

Avoidable rejection: the profile broke a stated preference (Smoking). The pre-send check would have blocked it.

Save to history

More screenshots, including full pages, are in [docs/screenshots/](#).

## Part 4: Curveball

**1. What I'd check first: is it used?** Compare the share of sent profiles that went through Match Check, and what happened to Blocks. A metric that doesn't move *at all* usually means the tool isn't in the workflow, not that the idea is wrong.

**2. Data I'd look at:** check logs joined to sent emails, split by matchmaker; override reasons; the share of checks returning "can't verify" (missing fields mean rules can't fire); 50 hand-labelled replies against the parser (if it mislabels, the metric is wrong); the rejection-reason mix before and after. Also sample size: two weeks is  $\approx 467$  profiles, where a 5-point rise in acceptance has  $p \approx 0.06$  and meetings lag by weeks, so I'd judge on the avoidable rate.

### 3. Iterate, change or kill?

- Not used → iterate on workflow: put the check inside the email step.
- Used, avoidable rate flat → preferences are captured badly at onboarding: fix intake.
- Avoidable rate fell, acceptance didn't → it worked, but the bottleneck is elsewhere. Keep it and target the next-largest reason.
- Kill only if it is used, measured correctly and still flat after four weeks.

## AI usage

- **Tools:** Claude Code
- **Used for:** turning my Part 1 notes into the final write-up, checking the funnel maths in a Python script, drafting Parts 2–4, generating mock data, and writing the prototype and its tests.

## Appendix: numbers and assumptions

| Stage                  | Count | Step conversion | % of shared | Lost at step |
|------------------------|-------|-----------------|-------------|--------------|
| Profiles shared        | 1,000 |                 | 100%        |              |
| Profiles accepted      | 310   | 31.0%           | 31.0%       | 690          |
| Contact details shared | 210   | 67.7%           | 21.0%       | 100          |
| Conversations started  | 150   | 71.4%           | 15.0%       | 60           |
| Meetings fixed         | 75    | 50.0%           | 7.5%        | 75           |
| Meetings completed     | 42    | 56.0%           | 4.2%        | 33           |

  

| Derived figure                         | Value                          | Assumption                                               |
|----------------------------------------|--------------------------------|----------------------------------------------------------|
| Avoidable rejections                   | ≈241 a month (24.1% of shared) | 35% applies to all 690 rejections                        |
| Acceptance excluding avoidable         | 310 / 758 = 40.9%              |                                                          |
| Profiles sent by A / B                 | ≈435 / ≈565                    | A and B send all 1,000 profiles                          |
| If B accepted at A's rate              | 440 accepted (+130)            | same client mix                                          |
| Projected acceptance                   | ≈41% (conservative 38.5%)      | blocked slots refilled at 40.9% (or 31%)                 |
| Projected meetings completed           | ≈55 a month (conservative 52)  | 13.5% of accepted profiles complete a meeting, unchanged |
| Search hours per month                 | ≈857                           | 100 active clients (not given)                           |
| Detecting +5 pts acceptance in 2 weeks | $z = 1.88, p = 0.06$           | ≈467 profiles vs a 1,000-profile baseline                |